APS110H1f2026 - October 2, 2026

October 1, 2026 11:58 AM

- Properties of Matter(s) - today again (and also for this week as well again) - also - "I Need More Structure in My Life" plus "Even More Structure"
  - Nano structure - bonding
  - Other nano structures to consider

Bonding - again

October 2, 2026 7:04 AM

status status.

Zn S

2+ 2⁻

cations anions

cationics anionics

re language

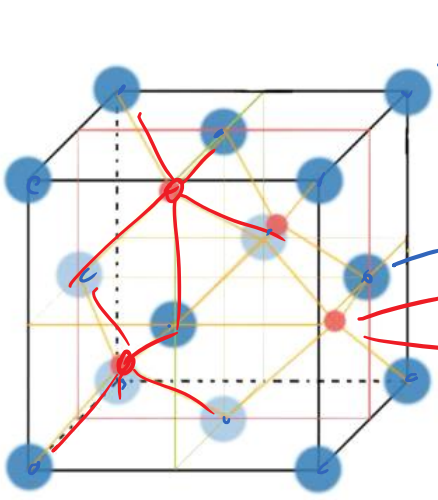


Figure 4. The complete ZnS ceramic crystal structure. I have drawn in the orange lines showing the direction of contact between cations and nearest neighbour anions.

ZnS

Zinc Sulfur

Hexagonal

$$V_{volume} = \frac{3\sqrt{3}}{2} a^2 c$$

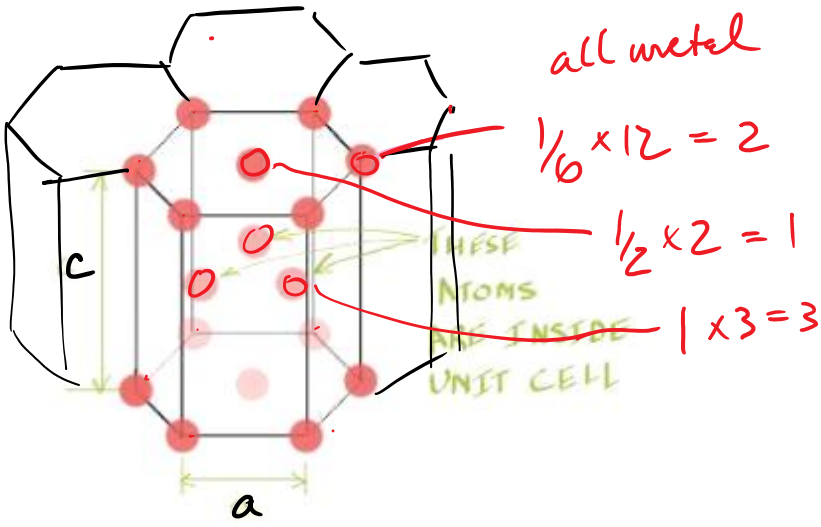
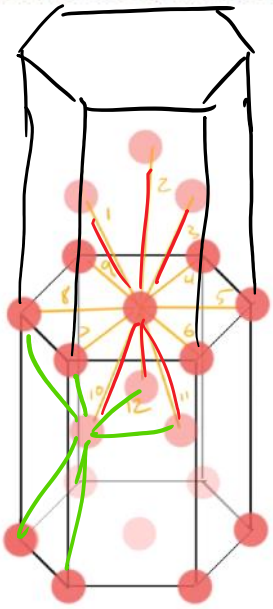


Figure 11: A hexagonal close packed unit cell. Note that the three atoms in the middle are inside the unit cell and are part of a close packed plane that is identical to the top and bottom planes.



Alternative to Consider Maybe - Nano Structure - 3

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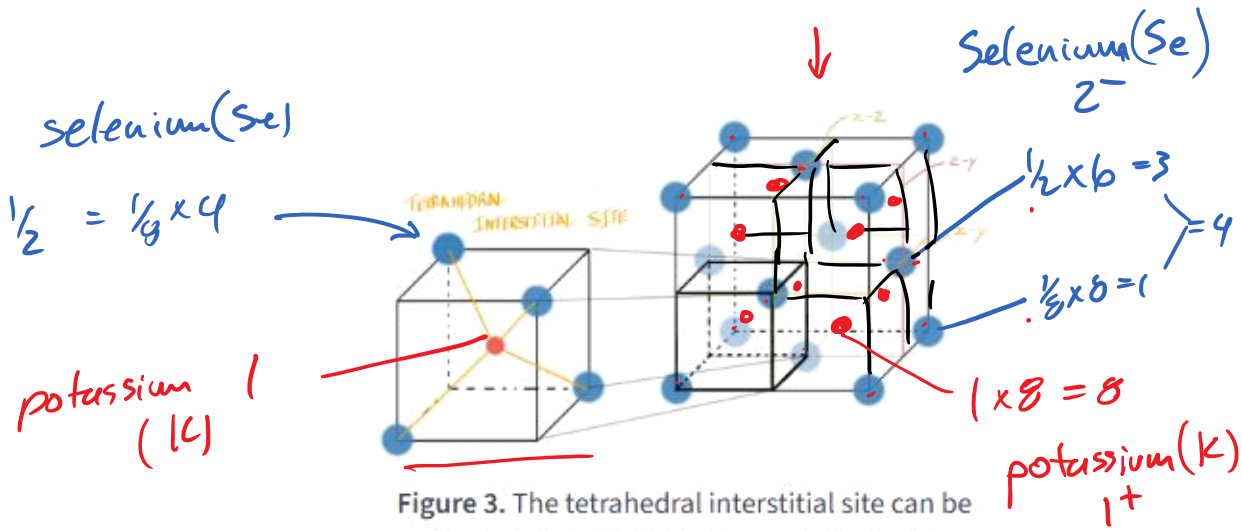


Figure 3. The tetrahedral interstitial site can be visualized as the centre of one of these sub-cubes. A cation occupying one of these sites would have a coordination number of 4.

you as an engineer:

you can choose ---

with unit to show the data.

- also a maybe - either is all valid.

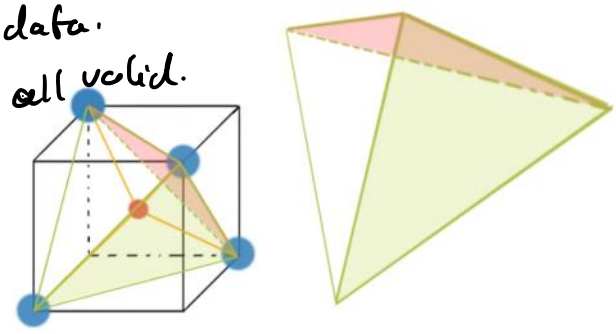


Figure 4. The tetrahedral interstitial site is so-named because it creates a four sided solid. In this illustration I have shaded in a couple of the faces to help visualize this geometric solid. In the tetrahedron drawn outside of the cube I also rotated it a little in the hopes that the perspective would be easier to visualize.